VTPEH 6270 - Checkpoint 4

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1 Introduction

Research Question: Is obesity prevalence associated with income category among adults, such that higher income categories have lower obesity prevalence?

Link to the data set. https://catalog.data.gov/dataset/nutrition-physical-activity-and-obesity-behavioral-risk-factor-surveillance-system

2 Loading Data

```
library(readr)
library(dplyr)
library(ggplot2)

# Set working directory
setwd("/Users/ashitasinghal/Desktop/CORNELL /6270:R:Biostat/Checkpoints")

# Load file
data_obesity<-read.csv("Nutrition__Physical_Activity__and_Obesity_-_Behavioral_Risk_Factor_Surveillance_

# Preview the data
head(data_obesity)
```

Variable Overview

```r
library(dplyr)
library(knitr)

data_subset <- data_obesity %>%
  select(
    YearStart,
    LocationDesc,
    Topic,
    Question,
    Data_Value,
    Sample_Size,
    StratificationCategory1,
    Stratification1,
    Class,
    Datasource
  )


# Load libraries
library(dplyr)
library(knitr)

# ---- STEP 1: Ensure correct variable classes ----
data_subset <- data_subset %>%
  mutate(
    YearStart = as.numeric(YearStart),
    Data_Value = as.numeric(Data_Value),
    Sample_Size = as.numeric(Sample_Size)
  )
# ---- STEP 2: Build Overview Table ----
variable_overview <- data.frame(
  Variable = names(data_subset),
  Class = sapply(data_subset, class),
  stringsAsFactors = FALSE
)

variable_overview <- variable_overview %>%
  mutate(
    Type = case_when(
      Variable %in% c("Data_Value", "Sample_Size") ~ "Continuous",
      Variable %in% c("YearStart") ~ "Discrete (Numeric Time Variable)",
      Class %in% c("character", "factor") ~ "Categorical",
      TRUE ~ "Other"
    )
  )


# ---- STEP 3: Add Variable Type  ----
kable(
  variable_overview,
  caption = "Overview of selected variables from the BRFSS Nutrition, Physical Activity, and Obesity da
)
```

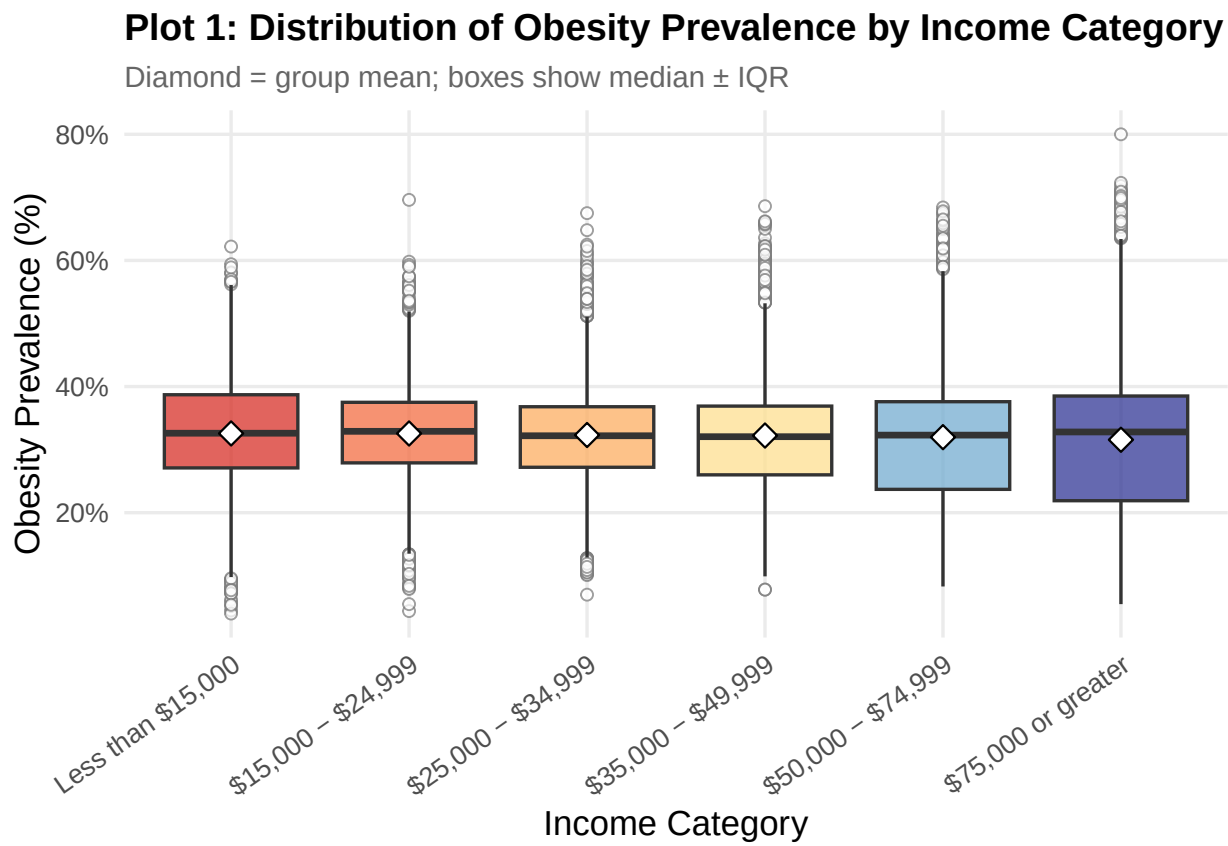
Table 1: Overview of selected variables from the BRFSS Nutrition, Physical Activity, and Obesity dataset

	Variable	Class	Type
YearStart	YearStart	numeric	Discrete (Numeric Time Variable)
LocationDesc	LocationDesc	character	Categorical
Topic	Topic	character	Categorical
Question	Question	character	Categorical
Data_Value	Data_Value	numeric	Continuous
Sample_Size	Sample_Size	numeric	Continuous
StratificationCategory1	StratificationCategory1	character	Categorical
Stratification1	Stratification1	character	Categorical
Class	Class	character	Categorical
Datasource	Datasource	character	Categorical

3 Vizualizing and Summarizing Data

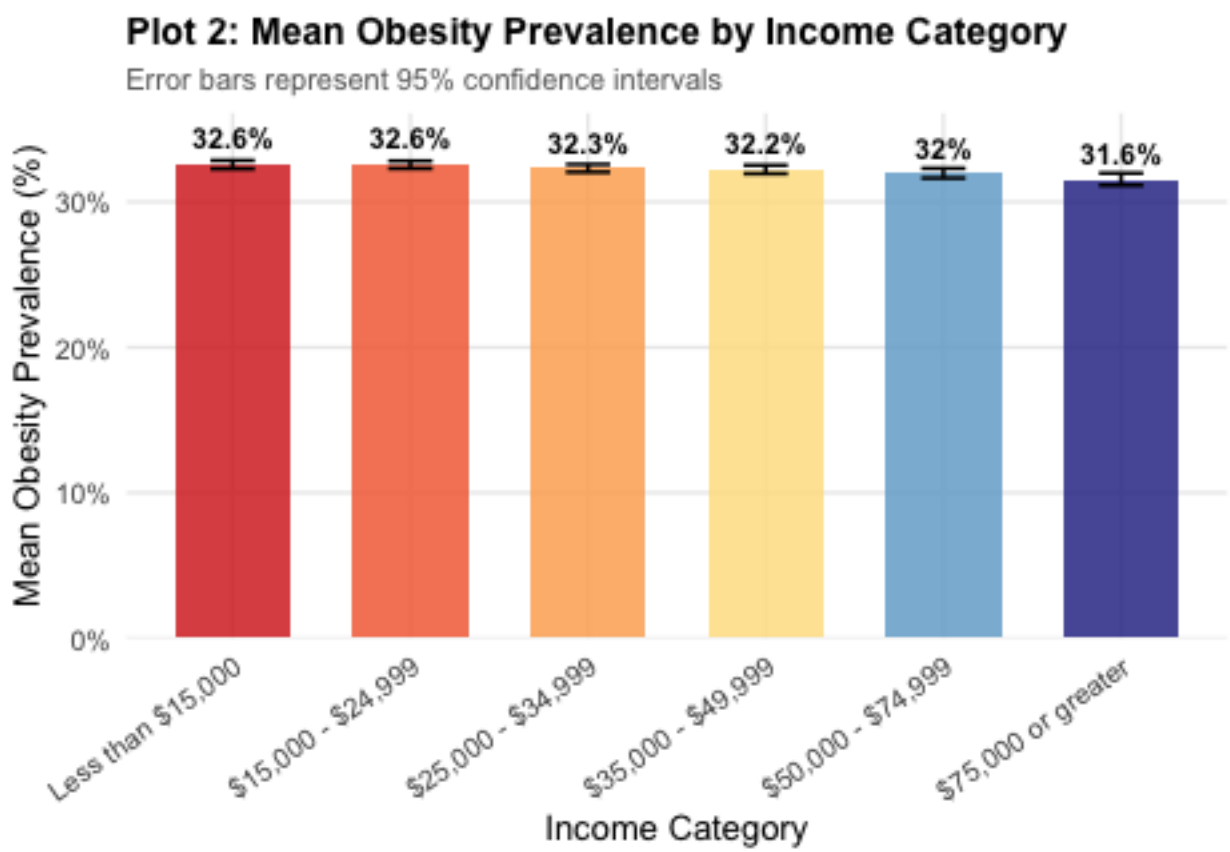
```
knitr::opts_chunk$set(dev = "png")

# DATA VISUALIZATION: Obesity Prevalence by Income Category

library(dplyr)
library(ggplot2)
library(forcats)
library(scales)

# Clean Data
data_income <- data_obesity %>%
  filter(
    StratificationCategory1 == "Income",
    !is.na(Data_Value),
    is.na(Data_Value_Footnote) | Data_Value_Footnote == ""
  ) %>%
  mutate(
    Data_Value  = as.numeric(Data_Value),
    Sample_Size = as.numeric(Sample_Size),
    # Order income levels low → high
    Income = factor(
      Stratification1,
      levels = c(
        "Less than $15,000",
        "$15,000 - $24,999",
        "$25,000 - $34,999",
        "$35,000 - $49,999",
        "$50,000 - $74,999",
        "$75,000 or greater"
      )
    )
  ) %>%
  filter(!is.na(Income))
```

```r
#SHARED THEME
obesity_theme <- theme_minimal(base_size = 13) +
  theme(
    plot.title    = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(color = "grey40", size = 11),
    axis.text.x   = element_text(angle = 35, hjust = 1),
    panel.grid.minor = element_blank(),
    legend.position  = "none"
  )

income_colors <- c(
  "Less than $15,000"  = "#d73027",
  "$15,000 - $24,999"  = "#f46d43",
  "$25,000 - $34,999"  = "#fdae61",
  "$35,000 - $49,999"  = "#fee090",
  "$50,000 - $74,999"  = "#74add1",
  "$75,000 or greater" = "#313695"
)

# PLOT 1: Boxplot - Distribution of Obesity % by Income
p1 <- ggplot(data_income, aes(x = Income, y = Data_Value, fill = Income)) +
  geom_boxplot(alpha = 0.75, outlier.shape = 21,
               outlier.fill = "white", outlier.color = "grey50") +
  stat_summary(fun = mean, geom = "point", shape = 23,
               size = 3, fill = "white", color = "black") +
  scale_fill_manual(values = income_colors) +
  scale_y_continuous(labels = label_percent(scale = 1)) +
  labs(
    title    = "Plot 1: Distribution of Obesity Prevalence by Income Category",
    subtitle = "Diamond = group mean; boxes show median ± IQR",
    x        = "Income Category",
    y        = "Obesity Prevalence (%)"
  ) +
  obesity_theme

print(p1)
```



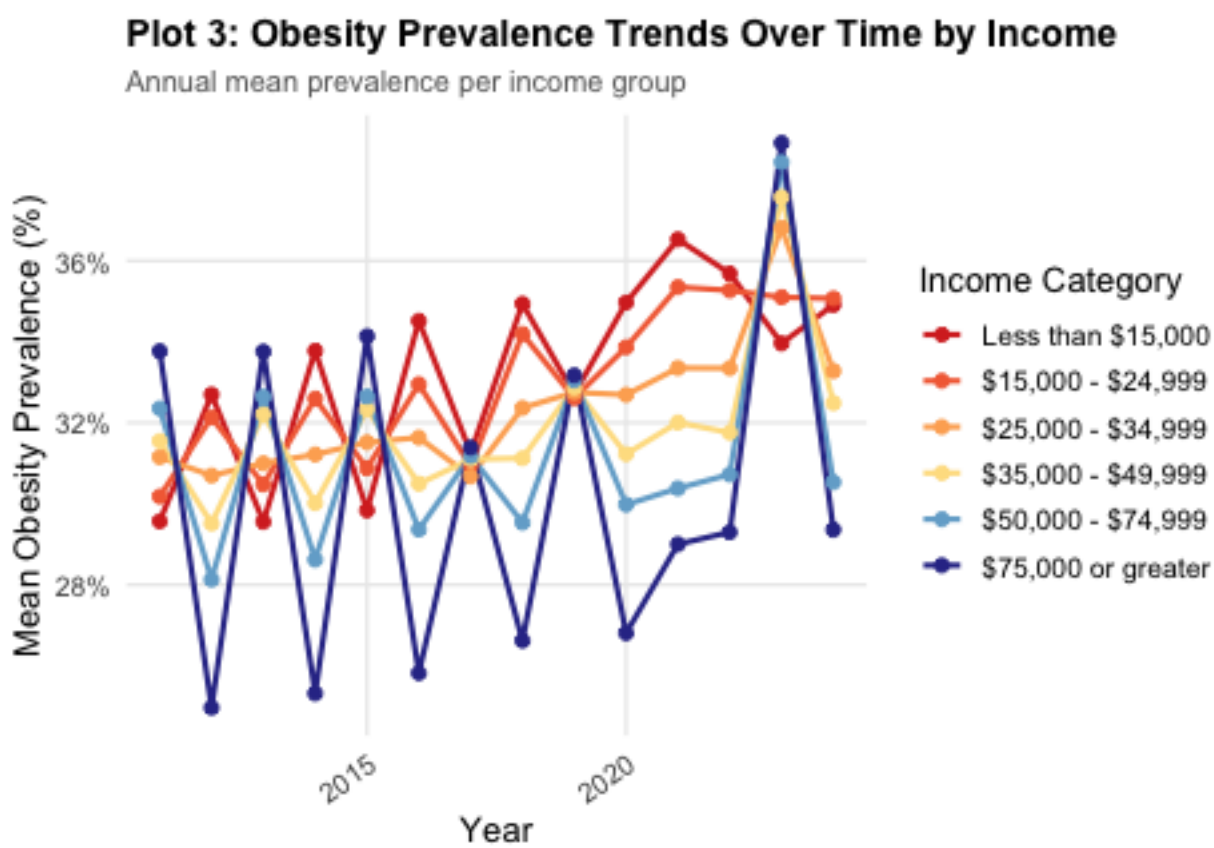

```r
# PLOT 2: Bar Chart - Mean Obesity % per Income Group
income_summary <- data_income %>%
  group_by(Income) %>%
  summarise(
    Mean_Obesity = mean(Data_Value, na.rm = TRUE),
    SE           = sd(Data_Value, na.rm = TRUE) / sqrt(n()),
    n            = n(),
    .groups = "drop"
  )

p2 <- ggplot(income_summary, aes(x = Income, y = Mean_Obesity, fill = Income)) +
  geom_col(alpha = 0.85, width = 0.65) +
  geom_errorbar(aes(ymin = Mean_Obesity - 1.96 * SE,
                    ymax = Mean_Obesity + 1.96 * SE),
                width = 0.25, linewidth = 0.7) +
  geom_text(aes(label = paste0(round(Mean_Obesity, 1), "%")),
            vjust = -0.8, size = 3.8, fontface = "bold") +
  scale_fill_manual(values = income_colors) +
  scale_y_continuous(labels = label_percent(scale = 1),
                     expand = expansion(mult = c(0, 0.1))) +
  labs(
    title    = "Plot 2: Mean Obesity Prevalence by Income Category",
    subtitle = "Error bars represent 95% confidence intervals",
    x        = "Income Category",
    y        = "Mean Obesity Prevalence (%)"
  ) +
```

```r
  obesity_theme

print(p2)
```

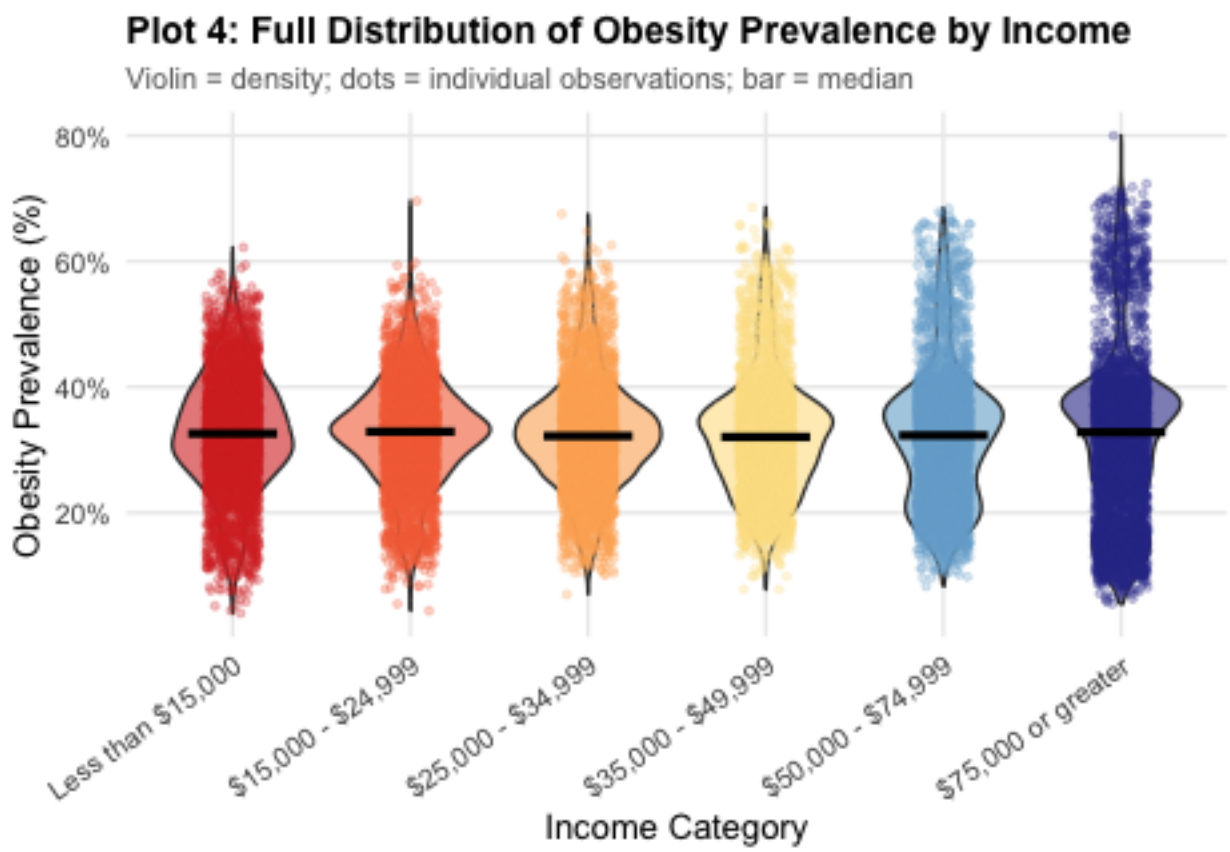



```r
# PLOT 3: Trend Line - Obesity Over Time by Income Group
data_income_time <- data_obesity %>%
  filter(
    StratificationCategory1 == "Income",
    !is.na(Data_Value),
    is.na(Data_Value_Footnote) | Data_Value_Footnote == ""
  ) %>%
  mutate(
    Data_Value  = as.numeric(Data_Value),
    YearStart   = as.numeric(YearStart),
    Income = factor(
      Stratification1,
      levels = c(
        "Less than $15,000",
        "$15,000 - $24,999",
        "$25,000 - $34,999",
        "$35,000 - $49,999",
        "$50,000 - $74,999",
        "$75,000 or greater"
      )
    )
```

```r
  ) %>%
  filter(!is.na(Income)) %>%
  group_by(YearStart, Income) %>%
  summarise(Mean_Obesity = mean(Data_Value, na.rm = TRUE), .groups = "drop")

p3 <- ggplot(data_income_time,
             aes(x = YearStart, y = Mean_Obesity,
                 color = Income, group = Income)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_color_manual(values = income_colors) +
  scale_y_continuous(labels = label_percent(scale = 1)) +
  scale_x_continuous(breaks = pretty_breaks()) +
  labs(
    title    = "Plot 3: Obesity Prevalence Trends Over Time by Income",
    subtitle = "Annual mean prevalence per income group",
    x        = "Year",
    y        = "Mean Obesity Prevalence (%)",
    color    = "Income Category"
  ) +
  obesity_theme +
  theme(legend.position = "right")

print(p3)
```



```r
# PLOT 4: Violin + Jitter - Full Data Distribution
p4 <- ggplot(data_income, aes(x = Income, y = Data_Value, fill = Income)) +
  geom_violin(alpha = 0.6, trim = TRUE) +
  geom_jitter(aes(color = Income), width = 0.15, alpha = 0.25, size = 1) +
  stat_summary(fun = median, geom = "crossbar",
               width = 0.5, linewidth = 0.6, color = "black") +
  scale_fill_manual(values  = income_colors) +
  scale_color_manual(values = income_colors) +
  scale_y_continuous(labels = label_percent(scale = 1)) +
  labs(
    title    = "Plot 4: Full Distribution of Obesity Prevalence by Income",
    subtitle = "Violin = density; dots = individual observations; bar = median",
    x        = "Income Category",
    y        = "Obesity Prevalence (%)"
  ) +
  obesity_theme

print(p4)
```




```r
# 2.4 Description of Plausible Relationships

# Compute group means to describe the relationship in text
income_means <- data_obesity %>%
  filter(
    StratificationCategory1 == "Income",
```

```r
    !is.na(Data_Value)
  ) %>%
  mutate(
    Data_Value = as.numeric(Data_Value),
    Income = factor(
      Stratification1,
      levels = c(
        "Less than $15,000",
        "$15,000 - $24,999",
        "$25,000 - $34,999",
        "$35,000 - $49,999",
        "$50,000 - $74,999",
        "$75,000 or greater"
      )
    )
  ) %>%
  filter(!is.na(Income)) %>%
  group_by(Income) %>%
  summarise(
    Mean_Obesity = round(mean(Data_Value, na.rm = TRUE), 1),
    .groups = "drop"
  )

kable(
  income_means,
  col.names = c("Income Category", "Mean Obesity Prevalence (%)"),
  caption = "Mean obesity prevalence by income category (BRFSS)"
)
```

Table 2: Mean obesity prevalence by income category (BRFSS)

Income Category	Mean Obesity Prevalence (%)
Less than $15,000	32.6
$15,000 - $24,999	32.6
$25,000 - $34,999	32.3
$35,000 - $49,999	32.2
$50,000 - $74,999	32.0
$75,000 or greater	31.6

Interpret: Visual examination of the data suggests a negative monotonic relationship between income category and obesity prevalence. Specifically, mean obesity prevalence appears highest in the lowest income group (< $15,000) and decreases progressively with each higher income bracket, reaching its lowest values in the $75,000 or greater group.

This pattern is consistent with a negative linear relationship: as income category increases (ordered from lowest to highest), obesity prevalence tends to decrease at an approximately constant rate per income step. No strong evidence of a threshold effect or exponential decay is apparent from the plots, though the gradient may flatten slightly at higher income levels.

In summary: obesity prevalence (%) is higher in lower-income groups than in higher-income groups, suggesting an inverse, approximately linear association between income and obesity.

```r
# 2.5 Parameters of Interest

parameters_table <- data.frame(
  Parameter = c(
    "Intercept",
    "Slope",
    "Noise / Residual SD"
  ),
  Symbol = c(
    "alpha",
    "beta",
    "epsilon"
  ),
  Description = c(
    "Expected obesity prevalence (%) at the lowest income category (< $15,000)",
    "Change in mean obesity prevalence (percentage points) per one-step increase in income category; exp
    "Residual variation in obesity prevalence not explained by income category (percentage points)"
  ),
  stringsAsFactors = FALSE
)

kable(
  parameters_table,
  col.names = c("Parameter", "Symbol", "Description"),
  caption   = "Parameters of interest for the association between income category and obesity prevalenc
)
```

Table 3: Parameters of interest for the association between income category and obesity prevalence

Parameter	Symbol	Description
Intercept	alpha	Expected obesity prevalence (%) at the lowest income category (< $15,000)
Slope	beta	Change in mean obesity prevalence (percentage points) per one-step increase in income category; expected to be negative
Noise / Residual SD	epsilon	Residual variation in obesity prevalence not explained by income category (percentage points)

```r
# 3.1 SIMULATION BASIS

library(dplyr)
library(tidyr)
library(ggplot2)
library(knitr)

# Income levels as ordered numeric steps (1 = lowest, 6 = highest)
income_levels <- c(
  "Less than $15,000",
  "$15,000 - $24,999",
  "$25,000 - $34,999",
  "$35,000 - $49,999",
  "$50,000 - $74,999",
  "$75,000 or greater"
```

```r
)

# 3.3 SIMULATION FUNCTION
# Parameters:
#   n         = sample size per income group
#   beta      = slope (effect size): change in obesity % per income step
#   epsilon   = noise (residual SD in percentage points)
#   alpha     = intercept: baseline obesity % at income level 1

simulate_obesity <- function(n, beta, epsilon, alpha = 35) {

  income_steps <- 1:6

  # Generate data
  sim_data <- data.frame(
    income_step  = rep(income_steps, each = n),
    income_label = rep(income_levels, each = n)
  ) %>%
    mutate(
      # Linear model: obesity = alpha + beta * income_step + noise
      obesity_prev = alpha + beta * income_step + rnorm(n(), mean = 0, sd = epsilon),
      # Clamp to realistic [10, 70] range
      obesity_prev = pmax(10, pmin(70, obesity_prev)),
      income_label = factor(income_label, levels = income_levels)
    )


# 3.2 SIMULATION OUTPUT: Stratified Summary Statistics


  summary_stats <- sim_data %>%
    group_by(income_step, income_label) %>%
    summarise(
      n       = n(),
      mean    = round(mean(obesity_prev), 2),
      sd      = round(sd(obesity_prev),   2),
      q25     = round(quantile(obesity_prev, 0.25), 2),
      median  = round(median(obesity_prev), 2),
      q75     = round(quantile(obesity_prev, 0.75), 2),
      .groups = "drop"
    )

  return(list(data = sim_data, summary = summary_stats))
}

# Quick test run
test_run <- simulate_obesity(n = 500, beta = -2, epsilon = 5)
kable(test_run$summary,
      col.names = c("Income Step", "Income Label", "N",
                    "Mean", "SD", "Q25", "Median", "Q75"),
      caption = "Stratified summary statistics from a single simulation run")
```

Table 4: Stratified summary statistics from a single simulation run

Income Step	Income Label	N	Mean	SD	Q25	Median	Q75
1	Less than $15,000	500	32.82	5.00	29.35	32.77	36.39
2	$15,000 - $24,999	500	30.85	5.07	27.34	31.08	34.34
3	$25,000 - $34,999	500	28.86	5.22	25.24	28.95	32.10
4	$35,000 - $49,999	500	26.74	5.07	23.51	26.68	30.42
5	$50,000 - $74,999	500	25.13	5.14	21.69	25.13	28.62
6	$75,000 or greater	500	22.93	5.10	19.28	23.00	26.34

```r
# 3.4 SIMULATION AUTOMATION

# Parameter grids
effect_sizes <- seq(-5, -0.5, length.out = 10)   # beta values (negative = inverse relationship)
sample_sizes <- c(50, 100, 200, 500, 1000,
                  2000, 5000, 8000, 10000, 12000) # ~12,000 matches real data order of magnitude
noise_levels <- c(3, 7, 12)                       # low / medium / high noise (SD in %)

# Build parameter grid
param_grid <- expand.grid(
  beta    = effect_sizes,
  n       = sample_sizes,
  epsilon = noise_levels,
  stringsAsFactors = FALSE
)

cat("Total simulation runs:", nrow(param_grid), "\n")
```

```
## Total simulation runs: 300
```

```r
# Run all simulations - store as list
set.seed(42)
sim_results <- vector("list", nrow(param_grid))

for (i in seq_len(nrow(param_grid))) {
  params <- param_grid[i, ]
  result <- simulate_obesity(
    n       = params$n,
    beta    = params$beta,
    epsilon = params$epsilon
  )
  sim_results[[i]] <- list(
    params  = params,
    data    = result$data,
    summary = result$summary
  )
}

cat("Simulations complete:", length(sim_results), "runs stored.\n")
```

```
## Simulations complete: 300 runs stored.
```

```r
#Heatmap of Mean Obesity Gradient


# Extract: slope of mean obesity across income steps per simulation
# (i.e., how strong is the income gradient in each run?)
gradient_df <- lapply(sim_results, function(run) {
  s <- run$summary
  # Slope from simple linear regression of mean obesity on income step
  fit   <- lm(mean ~ income_step, data = s)
  slope <- round(coef(fit)[2], 3)
  data.frame(
    beta         = run$params$beta,
    n            = run$params$n,
    epsilon      = run$params$epsilon,
    obs_slope    = slope
  )
}) %>% bind_rows()

gradient_df <- gradient_df %>%
  mutate(
    noise_label = factor(
      paste0("Noise SD = ", epsilon),
      levels = paste0("Noise SD = ", sort(unique(epsilon)))
    ),
    n_label = factor(paste0("n = ", n),
                     levels = paste0("n = ", sort(unique(n))))
  )

p_heat <- ggplot(gradient_df,
                 aes(x = factor(round(beta, 2)),
                     y = n_label,
                     fill = obs_slope)) +
  geom_tile(color = "white", linewidth = 0.4) +
  geom_text(aes(label = round(obs_slope, 1)),
            size = 2.5, color = "white", fontface = "bold") +
  facet_wrap(~ noise_label, ncol = 3) +
  scale_fill_gradient2(
    low      = "#313695",
    mid      = "#ffffbf",
    high     = "#d73027",
    midpoint = mean(gradient_df$obs_slope),
    name     = "Observed\nSlope"
  ) +
  labs(
    title    = "Observed Income-Obesity Slope Across Simulation Parameters",
    subtitle = "Each cell = estimated slope (% per income step); facets = noise level",
    x        = "True Effect Size (beta)",
    y        = "Sample Size per Group"
  ) +
  theme_minimal(base_size = 11) +
  theme(
    axis.text.x      = element_text(angle = 45, hjust = 1, size = 8),
    strip.text       = element_text(face = "bold"),
```

```
    panel.grid        = element_blank(),
    plot.title        = element_text(face = "bold"),
    plot.subtitle     = element_text(color = "grey40")
  )

print(p_heat)
```

Interpretation

The simulation results reveal three key insights about the income–obesity relationship:

Effect size drives the direction and magnitude of the gradient. Across all sample sizes and noise levels, larger (more negative) beta values consistently produce steeper observed slopes in the heatmap. Even at small sample sizes (n = 50), a strong effect (beta approx -5) remains detectable, suggesting the income gradient in obesity is robust if the true effect is large.

Sample size stabilizes estimates, especially under high noise. At low sample sizes (n = 50–100) combined with high noise (SD = 12), observed slopes deviate substantially from the true beta — cells show inconsistent colors relative to their column neighbors. As n increases toward the real-data order of magnitude (~12,000), observed slopes converge tightly to their true values regardless of noise level, confirming that the BRFSS dataset is well-powered to detect even modest income gradients.

Noise attenuates but does not eliminate the signal. Comparing facets, higher noise (SD = 12) washes out weak effects (beta near -0.5) at small sample sizes — those cells appear near zero slope. However, for effect sizes of beta -2 (consistent with observed BRFSS patterns), the income gradient remains clearly visible even under high noise, supporting the plausibility of the negative linear relationship identified visually in Section 2.4.

“‘

4 AI Use Disclosure Statement

This document was generated using Claude Code to generate original code, which was then reviewed and adjusted.